

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 14

Status: active BA3 execution plan after BA3-T2 cross-baseline identity/lifecycle pressure testing.

Supersedes: Revision 13 only for forward execution state; R1-R13 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- BA2 relation/action vocabulary: CLOSED BY BA2-T4.
- BA3-T1 provenance/origin lower bound: COMPLETED / PROVISIONAL PASS.
- BAreferent and BAProposition: ACCEPTED first-class semantic identity families.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.

2. BA3 objective

BA3 defines the smallest derivation/provenance/authority mechanics that connect governed documentation to Base Analysis while preserving project authority in governed documentation and making change/revalidation reproducible.

3. BA3-T1 - provenance/origin lower bound

Status: COMPLETED / PROVISIONAL PASS WITH LOWER-BOUND CANDIDATE.

T1 requires independent provenance on both BA1 identity families, many-to-many source lineage, immutable baseline context, exact source locator, the three origin states and explicit derivation basis/rationale for derived meaning.

4. BA3-T2 - cross-baseline identity, staleness and lifecycle

Status: COMPLETED / PROVISIONAL PASS WITH IDENTITY-LIFECYCLE REFINEMENT.

T2 replayed the facial-access M1/M2/M3 mutations and the order-fulfillment internal/external inventory-authority counterfactual.

Main dispositions

BAReferent continuity	retain when reusable referent meaning survives
BAProposition continuity	retain only under normalized assertion equivalence
source revision == BA revise	REJECTED
BA continuity dispositions	RETAIN REPLACE RETIRE
BA-level REVISE	NOT REQUIRED
review state	PENDING_REVIEW ACCEPTED REJECTED
freshness	CURRENT STALE
accepted REPLACE	prior element SUPERSEDED
accepted RETIRE	prior element RETIRED
resolved diagnostic	RETIRE or REPLACE, never mutate origin to GROUNDED

BA1 / BA2 reopen

NOT TRIGGERED

Mutation summary

- M1 Ethernet -> Wi-Fi retains abstract connectivity but replaces the realization proposition.
- M2 external -> project-owned transport replaces responsibility polarity, may retire external consumption and can introduce a new transport branch without changing the consumer transfer/security meaning.
- M3 remote -> local recognition retires the remote transfer and its old local constraints rather than dragging them onto the new behavior.
- external/internal WMS pressure produces the same responsibility-boundary pattern in a distinct domain.

5. Why BA3 remains open

T2 still does not close:

- exact derivation-rule/rationale registry and reproducibility;
- effective governed context/dependency semantics sufficient for principled staleness propagation;
- source-to-analysis and analysis-to-source feedback lineage after accepted correction;
- full BA3 closure review.

These must be tested before BA4 begins.

6. BA3-T3 - derivation-rule reproducibility and change-impact lineage pressure test**NEXT / NOT EXECUTED BY THIS PLAN REVISION.**

BA3-T3 must attack whether the T1/T2 contracts are sufficient for reproducible derivation and localized change impact without inventing a general dependency graph.

It must pressure-test at least:

1. the facial M4 raw-capture -> opaque-reference mutation where sibling governed context changes FR-3.4 meaning;
2. provider-specific raw state -> governed result normalization in order/payment/WMS cases;
3. the minimum identity and content required of `derivationRuleRef` or an equivalent inspectable rationale;
4. whether derived elements can be rebuilt/reviewed deterministically enough from `derivationBasis + rule/rationale`;
5. how direct source changes, derivation-basis changes and participant replacement mark only potentially affected BA elements stale;
6. whether explicit effective-context/dependency links are required beyond source parentage;
7. how an analysis-produced corrective candidate points back to the governed source area without becoming source authority;
8. how an accepted governed correction reconnects to the next BA baseline without turning analysis output into project truth;
9. whether any result forces BA1/BA2 or the BA3-T1/T2 lower bound to reopen.

BA3-T3 guardrails

Do not:

- design BA4 projections;
- define formal STRIDE/STRIDE-AI overlay schemas;

- define AnalysisRecord/Common Finding;
- design ThreatForge implementation classes/database tables;
- turn `derivationRuleRef` into an unrestricted programming language;
- make arbitrary legacy-document extraction a requirement.

BA3-T3 exit condition

Produce a falsifiable derivation/change-impact candidate and identify whether a distinct BA3 closure review is still required. Do not close BA3 merely because one implementation strategy can encode the examples.

7. BA4 - projections

Only after BA3 closes, define derived human and method projections from the same accepted BA semantics. No view becomes project authority.

8. BA5 - lexical vocabulary and optional assistance

Only after BA3/BA4 boundaries are stable, define display labels, authoring synonyms and optional assistance. NLP/LLM proposals remain candidate-producing support.

9. BA6 - complete Base Analysis regression and closure

Regress the complete BA0-BA5 design against the closed corpora and at least one structurally different holdout. BA6 may reopen only the smallest earlier responsibility on a material counterexample.

10. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

11. Next authorized microstep

Only after the BA3-T2 package is reviewed, committed, pushed and remotely verified, execute:

BA3-T3 - derivation-rule reproducibility and change-impact lineage pressure test.

Do not start BA4, formal threat-method overlays, Common Finding schema or implementation work.